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Clinical Research Paper

Non-inferiority of 1-minute versus 2-minutes cryoanalgesia cycles during minimally invasive repair of pectus excavatum: A propensity score-matched analysis

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ABSTRACT

Background and aim: Thoracoscopic cryoanalgesia reduces opioid requirements and hospital stay after minimally invasive repair of pectus excavatum (MIRPE), but increases operative time. Since the optimal freeze duration remains undefined, our protocol changed from 2-min to 1-min cycles in December 2024. Thus, we aimed to compare the effectiveness of 1-min vs 2-min cryoanalgesia cycles in MIRPE patients.
Methods: Patients treated with 1-min cryoanalgesia were matched to 2-min controls using Propensity Score-Matched (nearest-neighbor method) to improve the reliability of the comparison. Thirty-four patients were matched for demographic characteristics, pectus severity indices, and surgical variables. Postoperative opioid use, expressed as oral morphine equivalents (OME), and length of stay were compared using the Wilcoxon test and a non-inferiority analysis, with thresholds set at 10 mg OME and 0.5 days.
Results: Cryoanalgesia time was significantly shorter in the 1-min group (20.0 min [IQR: 18; 21] vs 29.0 min [IQR: 27; 31]; $p < 0.001$). No significant differences were observed in median length of stay (1 day [IQR: 1; 2] vs 1 day [IQR: 1; 1], $p = 0.25$) or in opioid requirement (10 mg OME [IQR: 0; 20] vs 2.5 mg OME [IQR: 0; 5], $p = 0.21$). Non-inferiority was confirmed for length of stay, as the upper one-sided 95 % confidence limit remained below the 0.5-day margin. For opioid consumption, the observed median difference remained clinically small, although the upper confidence limit slightly exceeded the predefined 10-mg margin.
Conclusion: One-minute cryoanalgesia cycles represent an effective and efficient alternative to 2-min cycles in MIRPE, reducing operative time without increasing postoperative opioid use or length of stay. While non-inferiority was statistically confirmed for length of stay, opioid requirements showed clinically minimal differences that approached—but did not formally meet—the predefined non-inferiority margin.
Type of study: Retrospective comparative study.
Level of evidence: III.
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HIGHLIGHTS

- The minimum duration of each cryoanalgesia cycle required to achieve effective analgesia after MIRPE remains undefined.
- This is the first study to compare 1- vs 2-min cycles in two groups of patients using matched analysis to improve reliability of the comparison.
- One-minute cryoanalgesia provides comparable postoperative opioid requirements and length of stay to conventional 2-min cryoanalgesia after MIRPE.

1. Background

Intercostal nerve cryoanalgesia (cryo) is increasingly used as an effective strategy for postoperative pain management in patients undergoing minimally invasive repair of pectus excavatum

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(MIRPE), with clear advantages over traditional analgesic approaches [1,2].

To achieve effective postoperative pain management, there is general consensus on applying cryoanalgesia to 4 to 7 intercostal spaces bilaterally, from the third to the seventh, in most reports [3–5]. However, the minimum duration of cryoanalgesia needed to provide effective analgesia remains undefined [6,7].

In 2016, the first report of cryo during MIRPE showed a significant reduction in opioid use and hospital length of stay (LOS) compared to traditional epidural analgesia [8]. Since then, multiple studies have repeatedly confirmed these benefits in both pediatric and adult groups [1,2,9–12].

Nevertheless, despite its effectiveness, a major limitation of cryo is the extended operative time, which can increase by more than 30 min, mainly due to the duration of freezing cycles [11,13]. In 2023, a retrospective study reported using 1-min cryo cycles in patients undergoing MIRPE for the first time and compared their results with a previous cohort of patients receiving analgesia through paravertebral catheters [6]. The study showed better postoperative pain control with cryoanalgesia, along with shorter hospital stays and reduced opioid use compared to traditional analgesia. In 2024, another study compared 1-min versus 2-min cryo cycles applied to opposite hemithoraces within the same patient, finding no differences in postoperative pain levels or sensory disturbances during the first 14 days after surgery in a group of 33 patients. To date, however, no study has directly compared different patient groups undergoing MIRPE treated with 1- versus 2-min cryo cycles. At our institution, a standard 2-min cryo protocol has been routinely used for postoperative pain management after MIRPE since 2018 [11]. Starting at the end of 2024, a 1-min cryo protocol was implemented. The aim of this study was to compare both approaches and determine whether 1-min cryo is non-inferior to the standard 2-min protocol regarding postoperative opioid use and LOS in propensity-matched cohorts of patients undergoing MIRPE.

2. Materials and methods

A comparative analysis was conducted between patients undergoing MIRPE with cryo using 1- and 2-min cycles at a single institution from October 2018 to June 2025. From October 2018 to November 2024, patients received 2-min cryo cycles. The perioperative analgesic protocol was standardized and remained unchanged throughout this period. In December 2024, the protocol was modified to apply 1-min cryo cycles in all MIRPE cases.

Demographic characteristics, pectus severity indexes and other tomographic measurements, intraoperative details, and postoperative outcomes were collected. Patients treated with 1-min cryoanalgesia were matched to 2-min controls by Propensity Score Matching (nearest-neighbor method) to account for potential confounders and achieve balanced baseline characteristics. The primary outcomes analysed were cryoanalgesia time, length of hospital stay and opioid consumption, expressed as oral morphine equivalents (OME). The secondary endpoint was the comparison of the postoperative pain, assessed using the Visual Analogue Scale (VAS).

Regarding non inferior analysis, the predefined margins were set at 10 mg OME and 0.5 days for LOS. The margins were intentionally conservative to ensure that any observed differences between the two groups would be clinically negligible. The 0.5-day LOS margin was selected to represent a difference substantially smaller than a full hospital day, indicating that the two protocols would result in an essentially comparable postoperative course. For opioid use, the 10 mg OME margin corresponds to 50 mg of intravenous tramadol, which represents the smallest rescue dose routinely administered in our institution during hospitalization.

All patients signed a written informed consent for the anonymous use of data. This study obtained IRB approval by the Universidad Abierta Interamericana, Buenos Aires, Argentina (protocol number: 0e1101).

2.1. Surgical technique, cryoanalgesia, and pain management protocol

All procedures were performed by the same surgical team using a standardized multidisciplinary approach, as described previously, and no surgical changes has been performed throughout the study period [11]. Cryo was administered using the "Cryo-S Painless" system (® METRUM Cryoflex, Blizne Łaszczyńskiego, Stare Babice, Poland) after selective orotracheal intubation and sternal elevation, resulting in complete thoracic remodeling. A 4th intercostal space access in the anterior axillary line was used to enter the thoracic cavity. The thoracoscopic cryoprobe was applied bilaterally to the intercostal nerves from T3 to T7, 3 cm lateral to the sympathetic chain, achieving a target temperature of −60 to −78 °C.

A standardized perioperative analgesic protocol was employed as previously described, including pre- and postoperative pregabalin for one week, NSAIDs and rescue tramadol during hospitalization, and NSAIDs exclusively at discharge (Table 1). Opioids were not prescribed after patient discharge [11]. Postoperative pain intensity was assessed using the Visual Analogue Scale (VAS), provided to patients via a free cross-platform app during the three weeks postoperatively.

2.2. Opioid use and OME calculation

Opioid consumption during hospitalization was extracted from electronic hospital records, including all administered opioid medications. Total opioid use was then converted into oral morphine equivalents (OME) using standard conversion ratios provided by an online opioid conversion calculator, based on established equivalency tables [14].

2.3. Statistical analysis

Patients were stratified into two groups according to the intraoperative cryoanalgesia cycle duration: 1 min or 2 min per nerve. To reduce selection bias, 1:1 propensity score matching (PSM) was performed, with matching variables including age, sex, weight, Haller index, correction index, number of pectus bars implanted, and presence of bar crossing. A caliper width of 0.2 standard deviations of the logit of the propensity score was applied. The primary outcomes were postoperative opioid consumption (in OME) and postoperative length of hospital stay. Continuous variables were summarized as medians with interquartile range (IQR), and categorical variables as counts and percentages. Continuous variables were compared using the Wilcoxon test, and categorical variables were analysed with Fisher's exact test. Non-inferiority was evaluated using the paired differences between groups. For each primary outcome, a Hodges–Lehmann estimator and its one-sided 95 % confidence interval were computed. Non-inferiority was concluded when the upper confidence limit did not exceed the predefined clinical margins (10 mg for opioid consumption, 0.5 days for length of stay). Post-hoc non-inferiority power was also estimated for interpretability, given the fixed sample size inherent to matched analyses. Statistical analyses were conducted using R version

Table 1
Pain regime for patients undergoing MIRPE with thoracoscopic intercostal cryoanalgesia.

Drug	Duration	Dosage form	Dose	Frequency
Preoperative (7 days)				
Pregabalin[a]	First 3 days	Tablets (oral)	50 mg	Q8h
	Next 4 days	Tablets (oral)	75 mg	Q8h
Postoperative (inpatient)				
Acetaminophen	Inpatient	Intravenous	1 g	Q8h (alt. with diclofenac)
Diclofenac	Inpatient	Intravenous	50 mg	Q8h (alt. with acetaminophen)
Tramadol	Rescue	Intraveous or Tablets (oral)	50 mg	Maximum Q8h
Pregabalin[a]	Inpatient	Tablets (oral)	75 mg	Q8h
Postoperative (outpatient)				
Omeprazole	4 weeks	Tablets (oral)	40 mg	Q24h
Acetaminophen	3 weeks	Tablets (oral)	1 g	Q8h (alt. with ibuprofen)
Ibuprofen	2 weeks	Tablets (oral)	400 mg	Q8h (alt. with acetaminophen)
Diclofenac	Rescue	Tablets (oral)	50 mg	Maximum Q8h
Pregabalin[a]	7 days	Tablets (oral)	75 mg	Q8h

[a] Oral gabapentin may be used instead of pregabalin (preoperatively: 3 days 100 mg every 8 h, next 4 days 200 mg every 8 h; postoperatively: 100 mg every 8 h).

Table 2
Table 1 shows demographic, tomographic, and surgical variables chosen to match patients from the groups undergoing 1- and 2- minute cryoanalgesia. Median and interquartile range were employed for continuous variables, whereas categorical variables were analysed using frequencies and percentages. Comparisons were performed using the Mann–Whitney U-test.

	Group 1-min	Group 2-min
N	17	17
Male	12 (70.6 %)	14 (82.4 %)
Age at surgery (years)	15.0 (14.0; 16.0)	15.0 (14.0; 17.0)
Weight (kg)	51.0 (46.5; 65.0)	54.0 (47.0; 66.0)
Haller Index	4.7 (4.2; 6.3)	4.4 (4.0; 5.6)
Correction Index (%)	44.0 (20.7; 58.6)	47.7 (24.5; 59.5)
Number of bars (n): 2	7 (41.2 %)	10 (58.8 %)
3	10 (58.8 %)	7 (41.2 %)
Patients with crossed bars	15 (88.2 %)	14 (82.3 %)

2024.04.0 + 735 (R Foundation for Statistical Computing) in RStudio version 2024 (Posit), with significance set at $p < 0.05$.

3. Results

In the Group 2-min, 268 patients were included, whereas the Group 1-min consisted of 17 patients. Seventeen matched controls for all 1-min cases were selected using propensity-matching (Table 2).

The comparison between the two groups is presented in Table 3. Cryo time was significantly shorter in the 1-min group (Group 1-min: 20.0 min [IQR: 18; 21] vs Group 2-min: 29.0 min [IQR: 27; 31], p < 0.001). No difference was observed in total opioid requirements (Group 1-min: 10 mg [IQR: 0; 20] vs Group 2-min: 2.5 mg [IQR: 0; 5], $p = 0.21$), in OME/days, OME/kg and OME/kg/day between both groups. Length of hospital stay was the same in the two groups (Group 1-min: 1 day [IQR: 1; 2] vs Group 2-min: 1 day [IQR: 1; 1], $p = 0.25$). No cases of neuropathic symptoms (paraesthesia, neuropathic pain) were identified in either group.

Table 3
Outcome comparison between 34 matched patients undergoing 1-min versus 2-min cryoanalgesia during MIRPE. Abbrev.: OME (Oral Morphine Equivalents).

	Group 1-min	Group 2-min	p
Cryoanalgesia duration (min)	20.0 (18.0; 21.0)	29.0 (27.0; 31.0)	<0.001
Length of stay (days)	1.0 (1.0; 2.0)	1.0 (1.0; 1.0)	0.254
OME (mg)	10.0 (0.0; 20.0)	2.5 (0.0; 5.0)	0.212
OME/days	8.67 (0.0; 10.0)	1.25 (0.0; 5.0)	0.233
OME/kg	0.2 (0.0; 0.31)	0.05 (0.0; 0.11)	0.221
OME/kg/days	0.1 (0.0; 0.3)	0.03 (0.0; 0.09)	0.272

Non-inferiority analysis showed divergent results across outcomes. For opioid consumption, the difference in medians was 7.5 mg, with a one-sided 95 % confidence interval ranging from 5.0 to 15.0 mg. Although the observed difference was clinically small and close to the predefined 10-mg margin, the upper confidence limit exceeded this threshold; therefore, statistical non-inferiority for OME could not be formally demonstrated. In contrast, postoperative length of stay met the non-inferiority criterion, with a paired median difference of 0 days and an upper one-sided 95 % confidence limit (<0.001 days) well below the 0.5-day margin. Post-hoc non-inferiority power was 8 % for OME and 30 % for length of stay, consistent with the limited sample available after matching.

Regarding pain evaluation after MIRPE, Visual Analogue Scale (VAS) scores remained consistently below 3 in both groups from the second postoperative day onward (Fig. 1).

4. Discussion

This study is the first to compare 1-min and 2-min intercostal cryoanalgesia cycles in MIRPE using a matched analysis. One-

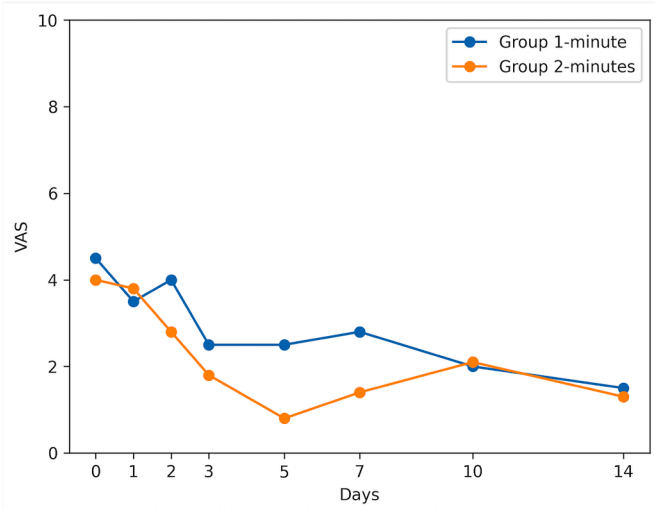

Fig. 1. Comparison of Pain Visual Analogue Scale between groups provided to patients with records of the three weeks postoperative. Note that scores remained under 3/10 after the second postoperative day in both groups. Abbrev.: VAS (Visual Analogue Scale of Pain).

minute cycles demonstrated equivalent clinical performance, with postoperative opioid requirements and hospital LOS remaining similar between groups. Non-inferiority was statistically confirmed for LOS, while opioid use showed clinically small differences that approached—but did not formally meet—the predefined non-inferiority margin. Also, while total cryo duration was significantly reduced in the 1-min group, no difference in postoperative pain perception was observed, as determined by patient-graded visual analogue scale.

Shortening cryo cycles to 1 min each reduced operative time by approximately 10 min in our cohort. Since prolonged operative time is a barrier to cryoanalgesia implementation, performing 1-min cycles might offer a more efficient alternative while maintaining safety and analgesic effectiveness.

The analgesic effects of intercostal nerve cryoablation in MIRPE were first reported by Keller et al. in 2016, who compared intraoperative cryoablation to thoracic epidural analgesia in 52 patients. They demonstrated significant reductions in opioid use and hospital stay, but an approximately 20-min increase in operative time [8]. Subsequent studies, including the randomized trial by Graves et al., confirmed markedly reduced opioid requirements and shorter hospitalization compared with epidural analgesia, but highlighted the issue of prolonged operative duration, reporting increases of up to 60–70 min [1,15]. Recently, the COPPER trial conducted by Bastianello et al. (2025) quantified the effect of standard 2-min cryo cycles on total procedural time, showing a 15-min increase compared to standard analgesia. In their randomized, single-center study of 88 patients, the total procedure duration for the cryoanalgesia group was 234 ± 25 min compared with 219 ± 31 min in the standard analgesia group [13]. Similarly, a systematic review by Eldredge et al. reinforced these findings, concluding that cryoanalgesia reduces LOS and opioid use but increases intraoperative cost, partly due to longer procedural times [9].

Histological studies provide a rationale for shorter cryo durations. Moorjani et al. demonstrated that both 1- and 2-min cryo nerve applications induce immediate axonal and Wallerian degeneration distal to the cryoprobe site, which interrupts nerve conduction and produces analgesia [16]. Importantly, both approaches similarly preserve the connective tissue sheaths, allowing axonal regeneration. Nerves treated with 1-min cycles showed axonal recovery after 1 week, while 2-min cycles showed delayed recovery starting at 2 weeks [16].

Clinical studies exploring shortened cryo cycles are limited. Zeineddin et al. were the first to evaluate 1-min cryoablation in a retrospective series of 198 patients, reporting shorter LOS and reduced opioid consumption than historical controls treated with paravertebral catheters [6]. More recently, Zacha et al. compared 1-min vs 2-min cycles in opposite hemithoraces of 33 patients, finding no differences in acute pain or sensory disturbances, supporting clinical equivalence [7]. However, no previous study has directly compared separate groups of patients undergoing MIRPE with 1-min versus 2-min cycles.

Our results confirm and extend previous findings. In our propensity-matched cohort of 34 patients, no significant differences were observed between the 1-min and 2-min cryo groups with respect to opioid consumption or hospital LOS. Although the 1-min cohort exhibited greater variability in opioid requirements, the median paired difference remained clinically small. The upper confidence limit, however, slightly exceeded the predefined 10-mg margin, preventing formal statistical confirmation of non-inferiority for OME. In contrast, LOS met the non-inferiority criterion. Postoperatively, ambulatory pain scores (VAS) were also comparable across most time points (Fig. 1). These findings suggest that reducing cryo duration from 2 to 1 min remains clinically effective while addressing its main procedural limitation —prolonged operative time.

Limitations of our study include its retrospective single-center design and a relatively small 1-min cohort size. Although residual confounding cannot be ruled out, propensity score matching was used to reduce bias. As expected in matched analyses, the fixed sample size limited the statistical power of the non-inferiority evaluation, particularly for OME. Future research is needed to confirm these findings and to examine long-term effects.

5. Conclusion

One-minute Cryoanalgesia cycles proved to be an effective alternative to the traditional 2-min cycles for managing postoperative pain after MIRPE. Reducing the freeze time significantly reduced operative duration without increasing opioid use, pain scores, or hospital stay. Overall, these results support adopting shorter cryo cycles as a more efficient strategy and with comparable outcomes.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Chat GPT 5.0 to improve language and readability. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

References

[1] Graves CE, Moyer J, Zobel MJ, Mora R, Smith D, O'Day M, et al. Intraoperative intercostal nerve cryoablation during the Nuss procedure reduces length of stay and opioid requirement: a randomized clinical trial. J Pediatr Surg 2019;54:2250–6. https://doi.org/10.1016/j.jpedsurg.2019.02.057.

[2] Lai K, Notrica DM, McMahon LE, Kang P, Molitor MS, Egan JC, et al. Cryoablation in 350 Nuss procedures: evolution of hospital length of stay and opioid use. J Pediatr Surg 2023;58:1435–9. https://doi.org/10.1016/j.jpedsurg.2022.10.051.

[3] Daemen JHT, De Loos ER, Vissers YLJ, Bakens MJAM, Maessen JG, Hulsewé KWE. Intercostal nerve cryoablation versus thoracic epidural for postoperative analgesia following pectus excavatum repair: a systematic review and meta-analysis. Interact Cardiovasc Thorac Surg 2020;31:486–98. https://doi.org/10.1093/icvts/ivaa151.

[4] Fraser JA, Briggs KB, Svetanoff WJ, Aguayo P, Juang D, Fraser JD, et al. Short and long term outcomes of using cryoablation for postoperative pain control in patients after pectus excavatum repair. J Pediatr Surg 2022;57:1050–5. https://doi.org/10.1016/j.jpedsurg.2022.01.051.

[5] Parrado R, Lee J, McMahon LE, Clay C, Powell J, Kang P, et al. The use of cryoanalgesia in minimally invasive repair of pectus excavatum: lessons learned. J Laparoendosc Adv Surg Tech 2019;29:1244–51. https://doi.org/10.1089/lap.2019.0203.

[6] Zeineddin S, Goldstein SD, Linton S, DeBoer C, Alayleh A, Ortiz I, et al. Effectiveness of one minute per level intercostal nerve cryoablation for postoperative analgesia after surgical correction of pectus excavatum. J Pediatr Surg 2023;58:34–40. https://doi.org/10.1016/j.jpedsurg.2022.09.032.

[7] Zacha S, Lopez M, Bilas J, Skonieczna-Żydecka K, Miegoń J, Biernawska J. One-versus two-minute intercostal nerve cryoanalgesia in children undergoing surgery for funnel chest deformity. J Pers Med 2024;14. https://doi.org/10.3390/jpm14080875.

[8] Keller BA, Kabagambe SK, Becker JC, Chen YJ, Goodman LF, Clark-Wronski JM, et al. Intercostal nerve cryoablation versus thoracic epidural catheters for postoperative analgesia following pectus excavatum repair: preliminary outcomes in twenty-six cryoablation patients. J Pediatr Surg 2016;51:2033–8. https://doi.org/10.1016/j.jpedsurg.2016.09.034.

[9] Eldredge RS, McMahon L. Intercostal nerve cryoablation therapy for the repair of pectus excavatum: a systematic review. Front Surg 2023;10. https://doi.org/10.3389/fsurg.2023.1235120.

[10] DiFiore JW, Robertson JO, Sung W. Sensory recovery following intercostal nerve cryoablation for the nuss procedure. J Pediatr Surg 2024:161904. https://doi.org/10.1016/j.jpedsurg.2024.161904.

[11] Toselli L, Gigena C, Bellia-Munzon G, Sanjurjo D, Vallee M, Martinez-Ferro M. Lessons learned after 176 patients treated with a standardized procedure of thoracoscopic cryoanalgesia during minimally invasive repair of pectus

excavatum. J Pediatr Surg 2024;59:372–8. https://doi.org/10.1016/j.jpedsurg.2023.10.047.

[12] Eldredge RS, Lai K, Ochoa B, Khoury E, Mihalcin K, Ostlie D, et al. A prospective study of sensory changes in pediatric patients after minimally invasive repair of pectus excavatum with cryoablation. J Pediatr Surg 2024;59:138–45. https://doi.org/10.1016/j.jpedsurg.2023.09.016.

[13] Bastianello M, Torre M, Bonfiglio R, Calevo MG, Palomba L, Uva P, et al. Cryoanalgesia for pain management after pectus excavatum repair (COPPER) in adolescents: a randomized controlled trial. Paediatr Anaesth 2025;35:347–58. https://doi.org/10.1111/pan.15090.

[14] Opiate Conversion Calculator n.d. https://www.mdcalc.com/calc/3947/opiate-conversion-calculator (accessed June 1, 2025).

[15] Perez Holguin RA, DeAngelo N, Sinha A, Shen C, Tsai AY. Cost and outcomes of intercostal nerve cryoablation versus thoracic epidural following the Nuss procedure. J Pediatr Surg 2023;58:608–12. https://doi.org/10.1016/j.jpedsurg.2022.12.011.

[16] Moorjani N, Zhao F, Tian Y, Liang C, Kaluba J, Maiwand MO. Effects of cryoanalgesia on post-thoracotomy pain and on the structure of intercostal nerves: a human prospective randomized trial and a histological study. Eur J Cardio Thorac Surg 2001;20(3):502–7.